

Machine Learning for Practical Quantum Error Mitigation [1]

Author: Haoran Liao et al.



Aniket Khan
ME21B021

Department of Mechanical Engineering
Indian Institute of Technology, Madras

May 15, 2025

Outline

1 Introduction & Motivation

2 ML-QEM Framework and Models

3 Experiments

- Performance Comparison at Tractable Scales
 - Random Circuits
 - Trotterized 1D Transverse-field Ising Model
- Mitigating Unseen Pauli Observables
- Enhancing Variational Algorithms
- Scalability through Mimicry

4 Discussion and Future Directions

Introduction and Motivation

- Quantum computing promises dramatic speedups over classical systems, potentially solving intractable problems.
- However, hardware noise severely limits performance.
- **Error Generation:**
 - *Decoherence:* Qubits interact with their environment, losing quantum information.
 - *Gate Imperfections:* Errors arise from imperfect control of quantum gates.
 - *Readout Errors:* Inaccuracies occur during the measurement process.
- Quantum error mitigation (QEM) aims to correct these errors without full fault-tolerance.
- Traditional QEM methods, such as Zero-Noise Extrapolation (ZNE), reduce errors at the expense of increased runtime and resource overhead.
- This work introduces a machine learning (ML) approach that predicts noise-free expectation values from noisy outputs, lowering overhead.

Quantum Error Mitigation (QEM) Overview

- **Traditional Methods:**

- **Digital Zero-Noise Extrapolation (ZNE):**

- Run circuit variants with artificially increased noise levels.
 - Extrapolate the results to estimate the noise-free (ideal) expectation value.

- **Probabilistic Error Cancellation (PEC):**

- Combine outcomes from circuits executed under different noise conditions.
 - Use probabilistic weights to cancel out errors.

- **Limitations:**

- High sampling overhead due to multiple circuit executions.
 - Increased runtime cost makes scaling challenging.

- **ML-QEM Idea:**

- Train statistical models (e.g., linear regression, random forests, MLPs, GNNs) to learn the mapping from noisy to near-ideal expectation values.
 - Leverage classical machine learning to mitigate errors with significantly lower overhead compared to traditional methods.

Why ML for Quantum Error Mitigation?

- **Modeling Complex Noise:** ML models capture nonlinear and high-dimensional relationships in noise that are hard to model with analytical methods.
- **Adaptability and Generalization:** ML can learn from diverse datasets, making it robust across various device noise profiles and circuit configurations.
- **Resource Efficiency:** Once trained, ML models mitigate errors without the need for additional circuit executions during runtime, lowering sampling overhead.
- **Scalability:** ML approaches can be scaled to large, intractable circuits where traditional methods become prohibitively expensive.
- **Adaptive Learning:** ML models can be fine-tuned to account for noise drift over time, maintaining accuracy as device characteristics change.

Overview of ML-QEM Framework

- **Workflow:**

- 1 Encode circuit-level features and noise parameters.
- 2 Obtain noisy expectation values from hardware/simulation.
- 3 Train an ML model to predict the mitigated (ideal) expectation values.

- **Advantages:** Reduced need for additional mitigation circuits; lower runtime overhead.

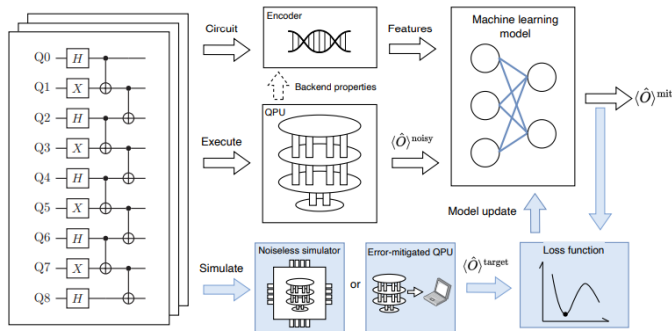
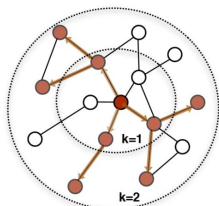


Figure: ML-QEM Workflow Diagram.

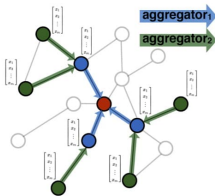
ML Models for QEM

The paper benchmarks four models:

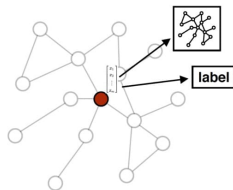
- **Linear Regression (OLS):** Simple, interpretable mapping using circuit and noise features.
- **Random Forest Regression (RF):** An ensemble of decision trees that consistently performs best.
- **Multi-Layer Perceptron (MLP):** Neural network with non-linear activation for complex error relationships.
- **Graph Neural Networks (GNN):** Encodes quantum circuits as graphs to capture both local and global information.



1. Sample neighborhood



2. Aggregate feature information from neighbors



3. Predict graph context and label using aggregated information

Figure: Graph Neural Network Image source: [2]

Linear Regression, Random Forest, and MLP Details

Linear Regression (OLS)

- Minimizes the squared error between predicted and ideal expectation values.

Random Forest Regression

- Ensemble of decision trees with bootstrap aggregating.
- Random feature selection helps reduce overfitting.
- Consistently outperforms other models in mitigating error while reducing computational overhead.

Multi-Layer Perceptron (MLP)

- Composed of fully connected layers with non-linear activations (e.g., ReLU).
- Trained using standard techniques such as the Adam optimizer with MSE loss.

Dataset Details:

- Features include the total number of gates and the frequency distribution of gate parameters.
- For example, gate rotation angles (e.g., 180°) are discretized into buckets, and the frequency of parameters in each bucket is used as a feature.

Graph Neural Networks (GNN)

- **Graph Representation:**

- Quantum circuits are converted into graphs where nodes encode quantum gate information.
- Edges represent qubit interactions; the circuit is directly transformed into graphical data, preserving edge relationships.

- **Node Features:**

- Each node is assigned a feature vector encoding the gate type, error parameters, and device-specific characteristics.

- **Message Passing and Aggregation:**

- Convolutional layers aggregate information from neighboring nodes and edges.
- This process captures both local and global correlations within the circuit.

- **Prediction:**

- The aggregated graph representation is fed into a fully connected layer for final prediction or classification.

- **Outcome:**

- This approach enables the GNN to effectively model the complex structure and noise characteristics of quantum circuits.

Overview of Experiments

- **A. Performance Comparison at Tractable Scales**
 - 1. Random Circuits
 - 2. Trotterized 1D Transverse-field Ising Model
- **B. Mitigating Unseen Pauli Observables**
- **C. Enhancing Variational Algorithms**
- **D. Scalability through Mimicry**

Performance Comparison at Tractable Scales

- The paper compares a diverse array of ML-QEM models to establish which methods are best suited for quantum error mitigation (QEM).
- Focus is on four statistical models:
 - Linear Regression with ordinary least squares (OLS)
 - Random Forest Regression (RF)
 - Multilayer Perceptrons (MLP)
 - Graph Neural Networks (GNN)
- Given the inherently non-linear relationship between noisy and ideal expectation values, non-linear models are particularly significant.
- Performance is evaluated on two classes of circuits:
 - **Random Circuits:** Unstructured circuits with varied two-qubit gate depths.
 - **Trotterized 1D Ising Spin Chain:** Structured circuits with distinct two-qubit gate arrangements.
- For Trotterized circuits, simulations are conducted under various noise conditions (incoherent, with/without readout error, and coherent noise) to assess both interpolation and extrapolation performance.

Experimental Setup: Random Circuits

- **Circuit Generation:**

- 4-qubit random circuits generated using Qiskit's random circuit generator.
- Two-qubit gate depths varied from 0 to 18 (step size of 2).
- Per depth: 500 training circuits and 200 test circuits per depth (total 2000 circuits).

- **Noise Model:**

- Simulations performed on IBM's FakeLima backend to emulate incoherent noise.
- Additional noise types (coherent, readout) can also be considered.

- **Error Metric:**

- Error defined as the L_2 distance between ideal and noisy single-qubit expectation values.

- **Benchmark:**

- Performance compared with digital Zero-Noise Extrapolation (ZNE).

Simulation Results: Random Circuits

• Results Overview:

- Error distributions (measured as the L_2 distance) are computed for each method across different gate depths.
- Bootstrap resampling (1,000 iterations) provides 95% confidence intervals.
- **Error Definition:** The error is defined as the L_2 distance between the vector of all ideal and noisy single-qubit expectations $\langle Z \rangle$, i.e., $\|\langle Z \rangle - \langle Z \rangle_{\text{ideal}}\|_2$, computed over $n = 2,000$ four-qubit random circuits.
- The Random Forest model consistently achieves lower errors compared to other ML models and ZNE.

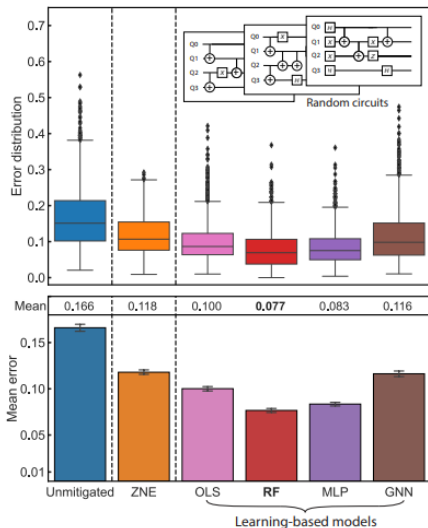


Figure: Comparison of error for various models

Trotterization and Circuit Construction

- **Hamiltonian: 1D Transverse-Field Ising Model**

$$\hat{H} = -J \sum_j \hat{Z}_j \hat{Z}_{j+1} + h \sum_j \hat{X}_j$$

- J is the coupling strength between neighboring qubits.
- h is the strength of the transverse magnetic field.
- $\hat{Z}_j$: Pauli Z operator acting on qubit j (eigenvalues $+1$ for 0 , -1 for 1).
- $\hat{X}_j$: Pauli X operator acting on qubit j (induces bit flips).
- The $-J \sum_j \hat{Z}_j \hat{Z}_{j+1}$ term represents interactions between adjacent spins, favoring alignment.
- The $h \sum_j \hat{X}_j$ term represents the effect of an external magnetic field causing spins to flip.

- **Trotterization:**

- Discretizes the continuous time evolution into multiple, small time steps.
- Each step is implemented via a quantum circuit, approximating the dynamics.

- **Circuit Construction:**

- Circuits are built with layers of CNOT gates (to simulate spin interactions) and single-qubit rotations (to simulate the magnetic field).
- Coupling strengths are uniformly sampled from the paramagnetic phase ($J < h$).

Experimental Setup: Trotterized Circuits

- **Data Generation:**

- For each Trotter step, 300 training and 300 test circuits are generated.
- Training circuits cover up to 14 Trotter steps; testing circuits extend up to 29 steps.

- **Measurement:**

- Each circuit is measured in a randomly chosen Pauli basis for all weight-one observables.

- **Noise Models:**

- Simulations on IBM's FakeLima backend emulate incoherent noise.
- Coherent errors are modeled as CNOT over-rotations, with an average over-rotation angle of approximately 0.04π .

- **Testing Scenarios:**

- Both interpolation (within the training range) and extrapolation (beyond the training range) are evaluated.

- **Practical Consideration:**

- In extrapolation, none of the models effectively mitigate errors.
- In practice, applying techniques like dynamical decoupling and randomized compiling can suppress coherent errors.
- This pre-correction converts coherent noise into incoherent noise, enabling ML-QEM methods to perform optimally.

Simulation Results: Trotterized Circuits

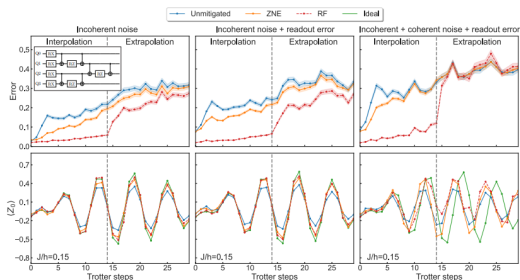


Figure: Mitigation accuracy under varying noise complexities and testing scenarios.

- **Top row:** Average L_2 error on Trotter circuits (4 CNOT layers) simulating a 1D TFIM, averaged over 300 circuits.
- Vertical dashed line separates ML-QEM interpolation (left) from extrapolation (right).
- Noise complexity increases from left to right, with coherent errors (e.g., CNOT over-rotations) added.
- **Bottom row:** Typical error-mitigated $\langle Z_0 \rangle$ values for $J/h = 0.15$

Hardware Results: Trotterized Circuits

QPU Hardware Validation:

- Experiments are conducted on IBM's ibm alqiers using up to 100-qubit Trotter circuits.
- For each Trotter step, 50 training circuits and 250 test circuits are executed.
- Each circuit is measured in a randomly chosen Pauli basis for weight-one observables.
- Results show that ML-QEM (using RF) achieves a 40% overall and 50% runtime overhead reduction compared to digital ZNE.
- Shaded regions in the plots denote $\pm$ standard error from bootstrap resampling.

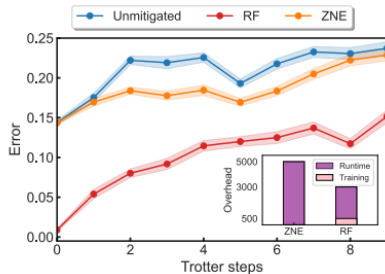


Figure: Average execution error on QPU hardware for different methods.

Experimental Setup: Mitigating Unseen Pauli Observables

Overview:

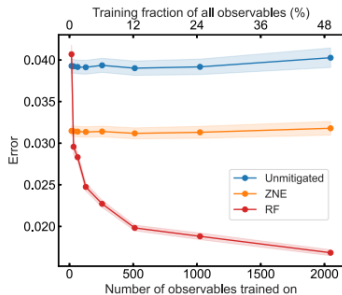
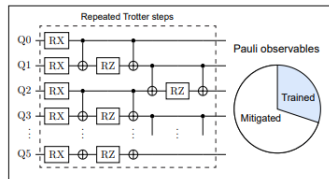
- Quantum routines (e.g., state tomography, VQE) require the measurement of many non-commuting Pauli observables.
- This results in multiple target circuits with identical gate sequences but different measurement bases, leading to high overhead.

Data Generation and Training:

- A Trotterized Ising circuit (6 qubits, 5 Trotter steps) with $J/h = 0.15$, producing up to $4^n - 1 = 4095$ distinct Pauli observables was used in the paper
- Training data was generated by using a small fraction (as low as $\lesssim 2\%$) of these observables.
- Then the ML model was applied to mitigate noisy expectation values from the rest of all possible Pauli observables.

Results: Mitigation of Unseen Pauli Observables

- **Simulation Results:** The RF model trained on $\lesssim 2\%$ of the 4095 observables achieves lower L_2 errors compared to digital ZNE.
- **Top right:** Circle depicts the pool of all possible $4n$ Pauli observables. Shadings depicts the fraction of observables used in training the ML model; the remaining observables are unseen prior to deployment in mitigation.
- **Bottom:** Average error of mitigated unseen Pauli observables versus the total number of distinct observables seen in training. The shaded regions represent



Performance for unseen Pauli Observables

Experimental Setup: Enhancing VQE with ML-QEM

Overview:

- The Variational Quantum Eigensolver (VQE) estimates the ground-state energy of a quantum system.
- It optimizes circuit parameters $\vec{\theta}$ iteratively to minimize the measured energy $\langle \hat{H} \rangle_{\vec{\theta}}$.
- Noisy execution distorts the measured energy, requiring error mitigation.

Mitigation Approach:

- Conventional QEM methods (e.g., ZNE) require additional mitigation circuits at each optimization step, increasing runtime.
- ML-QEM is trained in advance to correct noisy energy estimates without extra mitigation circuits per iteration.

Training Details:

- ML-QEM (RF model) is trained on 2,000 circuits with randomly sampled parameters $\vec{\theta} \in [-5, 5]$.
- A two-local variational ansatz circuit is used, with eight tunable parameters.

Results: ML-QEM Performance in VQE

- ML-QEM (RF) achieves energies close to chemical accuracy.
- Absolute errors from ML-QEM are smaller than those from ZNE-mitigated energies.
- No readout error mitigation was applied for this simulation
- The ML model generalizes well across different parameter values without prior knowledge of the noise model.

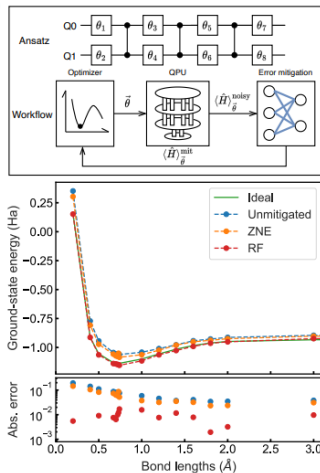


Figure: (Top) VQE ansatz circuit and optimization workflow. (Bottom) Ground-state energy estimates of H_2

Experimental Setup: Mimicry for Large-Scale Circuits

Large-scale circuits make classical simulation of ideal expectation values infeasible.

Experiment Details:

- A **100-qubit** Trotterized 1D TFIM circuit is executed on `ibm brisbane`.
- **ZNE with TREX** (Twirled Readout Error Extinction) is used as a reference mitigation method.
- ML-QEM is trained to mimic ZNE-mitigated values for **5 single-qubit** $\langle Z_i \rangle$ observables.
- Each Trotter step contains **4 layers of parallel CNOT gates**.
- At Trotter step **10**, the circuit contains 1,980 CNOT gates.

Benchmarking Approach:

- ZNE is first validated on a **Clifford 100-qubit Trotterized circuit** ($h = 0.5\pi, J = 0$), which allows classical simulation.
- ML-QEM is trained to match ZNE-mitigated values across varying **Trotter steps** and **J/h values**.
- **Pauli twirling** (5 twirls per circuit) is applied before either ZNE extrapolation or ML-QEM mitigation.

Results: Mimicry Performance of ML-QEM

Key Findings:

- **ML-QEM accurately mimics ZNE** across all Trotter steps.
- Distance between ZNE-mitigated and ML-QEM-mitigated values remains **low** for non-Clifford circuits.
- The L2 norm of errors between ZNE-mitigated and ML-QEM-mitigated values is consistently small.
- ML-QEM achieves a **25% reduction in overall quantum resources** and a **50% reduction in runtime overhead**.

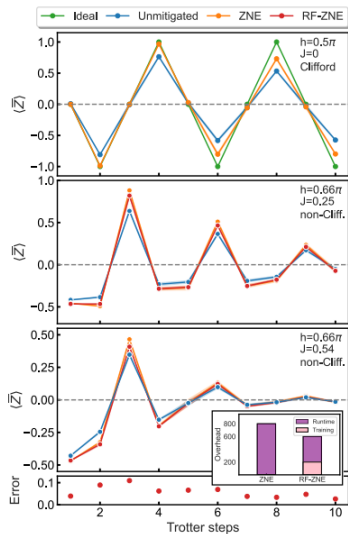


Figure: Comparison of ML-QEM-mitigated and ZNE-mitigated expectation values

Discussion and Future Work

- ML-QEM offers a promising path to lower overhead error mitigation on near-term devices.
- **Key Points:**
 - Efficiently reduces runtime overhead.
 - Can generalize across different circuit types and noise profiles.
- **Future Research:**
 - Improved encoding strategies (e.g., incorporating pulse shapes).
 - Adaptation to noise drift via fine-tuning.
 - Benchmarking against other advanced QEM techniques.

Conclusion

- ML-QEM leverages classical machine learning to significantly mitigate quantum errors.
- Demonstrated superior performance and efficiency compared to traditional methods.
- Provides a scalable approach to enhance quantum computations on noisy hardware.

Thank You!

References I

- [1] Haoran Liao, Derek S. Wang, Iskandar Sitdikov, Ciro Salcedo, Alireza Seif, and Zlatko K. Minev. Machine learning for practical quantum error mitigation. *Nature Machine Intelligence*, 2024.
- [2] NVIDIA. What are graph neural networks?
<https://blogs.nvidia.com/blog/what-are-graph-neural-networks/>, 2023.